

STANES SCHOOL – CBSE

Class 12 – Informatics Practices Cyclic Test-2

Marks: 25

Time: 1 hour

SECTION A - Objective Type Questions (5 x 1 = 5)

1. Which clause filters grouped records after GROUP BY? (1)
(a) WHERE (b) HAVING (c) AS (d) INNER JOIN
2. Select the query that correctly displays each CITY with its average MARKS under the alias AvgMarks. (1)
(a) SELECT CITY, AVG(MARKS) AS AvgMarks FROM STUDENT GROUP BY CITY;
(b) SELECT CITY, AVG(MARKS) FROM STUDENT HAVING CITY;
(c) SELECT CITY, AVG(MARKS) AS AvgMarks FROM STUDENT WHERE AVG(MARKS);
(d) SELECT AVG(MARKS), CITY FROM STUDENT;
3. Write the output of:
SELECT ROUND(POWER(3, 2) / 4, 1); (1)
4. Name the function that returns the position of the first occurrence of a substring within a string. (1)
5. Correct the following query so that only departments having more than three employees are displayed: (1)
SELECT DeptId, COUNT(*) FROM EMPLOYEE WHERE COUNT(*) > 3 GROUP BY DeptId;

SECTION B - Short Answer Questions (5 x 2 = 10)

6. Write SQL queries for the following: (2)
(a) Display EMPNAME in uppercase and the first three characters of CITY from EMPLOYEE.
(b) Display SALARY increased by 10% as RevisedSalary using aliasing.
7. Write SQL queries for the following: (2)
(a) Display the month name and day name of DOJ from EMPLOYEE.
(b) Display the date part of the current date and time.
8. State whether each statement is True or False. Rewrite the false statement correctly. (2)
(a) WHERE can contain aggregate functions to filter groups.
(b) HAVING is used with GROUP BY and may contain aggregate functions.
9. Write one SQL statement using GROUP BY and HAVING to display DeptId and the maximum Salary for departments whose average Salary is greater than 50000. (2)
10. Tables EMPLOYEE(EmpId, EmpName, DeptId) and DEPARTMENT(DeptId, DeptName) have the common column DeptId. Write: (2)
(a) an INNER JOIN query to display EmpName and DeptName; and
(b) a NATURAL JOIN query to display all matching rows.

SECTION C - Table-based SQL Question (10 x 1 = 10)

11. Consider the following tables EMPLOYEE and DEPARTMENT. Write SQL queries for parts (a)-(e) and write the outputs for parts (f)-(j).

(10)**Table: EMPLOYEE**

EmpId	EmpName	DeptId	City	Salary	DOJ
E01	Aarav	D01	Delhi	72000	2022-07-15
E02	Meera	D02	Mumbai	68000	2021-01-10
E03	Kabir	D01	Delhi	54000	2023-11-06
E04	Riya	D03	Pune	48000	2020-03-23
E05	Ishaan	D02	Mumbai	60000	2022-09-12
E06	Tara	D01	Noida	66000	2021-12-01

Table: DEPARTMENT

DeptId	DeptName
D01	Technology
D02	Finance
D03	Human Resources

Write SQL queries:

- (a) Display DeptId and average Salary as AvgSalary for each department.
- (b) Display DeptId and number of employees for departments having more than two employees.
- (c) Display EmpName, the first three letters of EmpName as Code, and LENGTH(EmpName) as NameLength.
- (d) Display EmpName, MONTHNAME(DOJ) as JoinMonth and YEAR(DOJ) as JoinYear.
- (e) Using an EQUI JOIN, display EmpName and DeptName for matching departments.

Write the output of each query:

- (f) `SELECT COUNT(*), COUNT(DISTINCT City) FROM EMPLOYEE;`
- (g) `SELECT DeptId, MAX(Salary) FROM EMPLOYEE GROUP BY DeptId;`
- (h) `SELECT UPPER(EmpName), RIGHT(City, 3) FROM EMPLOYEE WHERE EmpId = 'E04';`
- (i) `SELECT MOD(Salary, 7000), ROUND(Salary / 12, 2) FROM EMPLOYEE WHERE EmpId = 'E03';`
- (j) `SELECT EmpName, DeptName FROM EMPLOYEE INNER JOIN DEPARTMENT ON EMPLOYEE.DeptId = DEPARTMENT.DeptId WHERE EmpId = 'E05';`